RAJALAKSHMI ENGINEERING COLLEGE

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C2

Started on	Thursday, 20 August 2026, 10:32 AM
State	Finished
Completed on	Thursday, 20 August 2026, 10:35 AM
Time taken	2 mins 34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Function to find the majority element using Boyer-Moore Voting Algorithm
int majorityElement(int* nums, int numsSize) {
    int candidate = nums[0];
    int count = 0;

    for (int i = 0; i < numsSize; i++) {
        if (count == 0) {
            candidate = nums[i];
        }
        if (nums[i] == candidate) {
            count++;
        } else {
            count--;
        }
    }

    return candidate;
}

int main() {
    int n;

    // Read the size of the array
```

```
    while (scanf("%d", &n) != EOF) {
        int *nums = (int *)malloc(n * sizeof(int));
        if (nums == NULL) {
            return 1;
        }

        // Read the elements of the array
        for (int i = 0; i < n; i++) {
            scanf("%d", &nums[i]);
        }

        // Find and print the majority element
        int result = majorityElement(nums, n);
        printf("%d\n", result);

        free(nums);
    }

    return 0;
}
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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